

DutchBay 150 MW Wind Farm — Complete Feasibility & Recommendation Study

Project: DutchBay Wind Farm — 159.6 MW (15 × IEA-Reference-10 MW), Dutch Bay, Kalpitiya, Puttalam District, Sri Lanka **Prepared:** 2026-07-07 · **Finance re-baselined:** 2026-08-17 (F5-01, #1034/#1038) **Basis:** live pipeline run, DutchBay EPC model v15.3.1, origin/main 7e64d33 **Canonical scenario:** scenarios/dutchbay_lendercase_2025Q4.yaml (5th-generation canon) **Status of this document:** lender/DFI-grade feasibility. Every headline number traces to a fresh run artifact (§Appendix D). Framing is honest: on its committed flat-LKR PPA the project is **value-destructive**, and this study says so plainly. **Re-baseline scope (2026-08-17):** the F5-01 correction aligns the operating FX path to COD (see Appendix A). It moves the **finance** layer only — every finance figure below (\$0, \$7.1, \$7.2, \$7.5, Appendices A/B/D) was regenerated live at v15.3.1. The wind-resource, GeoGIS, micro-siting and grid-screen results are **inputs** to the finance layer, are unaffected by an FX-timing correction, and are carried forward unchanged from the 2026-07-07 full-stack run. **Run basis (full-stack):** unlike a finance-only pass — which reads a committed AEP summary — this study fired **every feasibility-relevant module end-to-end:** the wind-resource chain (ERA5 → Weibull → wake → bankable AEP), GeoGIS, micro-siting, the finance engine, a 100k-trial Monte-Carlo, global (Sobol/PAWN) sensitivity, the capital-structure optimiser, and the grid screen. The long path **reproduces the committed baseline** (fresh AEP 464.36 GWh → canon KPIs), and the exhaustive run **surfaced and fixed two real bugs** (grid boundary_clip #929, ride_through #930).

o. Executive summary & recommendation

DutchBay is a **technically sound, financially unviable-as-structured** 159.6 MW onshore wind project. The wind resource is real and bankable-grade; the economics fail because the revenue line — a **flat, non-escalating, non-FX-indexed 20.3 LKR/kWh CEB tariff** — cannot service USD/blended debt and clear an equity return once the LKR is allowed to depreciate at its data-derived ~5.9%/yr.

Headline economics (live, canonical lender case):

Metric	Value	Read
Project IRR (unlevered, real)	-0.12 %	far below the ~10.3 % build-up WACC → value-destructive
Equity IRR (levered)	-7.85 %	negative; equity MOIC 0.41 (< 1)
Project NPV @ WACC	-\$91.81 M	negative in every scenario run
Min DSCR (covenant / per-period)	1.30 / 1.30	0 equity-lockup years; sculpt holds 1.30 per period
LLCR / PLCR	1.27 / 1.31	thin but positive coverage
DSCR-solved gearing	0.355	debt auto-sized down to hold the 1.30 floor

The one-line verdict: *the turbines will spin and the debt can (just) be sculpted to a 1.30 DSCR, but at the administered flat-LKR tariff the project returns -0.12 % against a ~10.3 % cost of capital — it destroys value on a merchant/standalone basis.* The only way it clears is the **blended-finance / concessional case** (cf. the project's own Final EIA, which reports an **18.07 % economic IRR** under a very different capital-and-benefit frame).

Recommendation: Conditional NO-GO on the current commercial terms; conditional GO only under a restructured revenue or capital stack. Do not reach financial close at a flat-nominal-LKR 20.3 LKR/kWh tariff. Pursue, as conditions precedent, at least one of: (i) tariff FX-indexation or escalation; (ii) a materially concessional / grant-blended senior tranche; (iii) a shorter-tenor or availability-backed PPA that lifts the achievable DSCR. §11 lists the full conditions-precedent set. Development readiness rolls up to **RED** (financing uncommitted) despite a **GREEN** environmental status.

1. Project & context

Attribute	Value
Capacity	159.6 MW (15 × IEA Reference 10 MW, 10.638 MW rated)
Site	Dutch Bay, Kalpitiya, Puttalam District, North Western Province
Turbine-array centroid	8.27 °N, 79.75 °E
Grid POI	Puttalam 220 kV substation (33 kV collector → 220 kV)
COD	2027 (2-yr construction, 1-yr PPA-to-FC)
Operating life	20 years
Offtaker	Ceylon Electricity Board (CEB), credit B+ (S&P)
PPA	CEB standardized BOO, 20-yr, flat 20.3 LKR/kWh, fixed, no escalation, no FX index
Developer	DutchBay Wind Power Pvt Ltd

The 220 kV evacuation line is a **separate CEB-funded project** (not in this project's capex or wheeling charge). The site sits in the same Kalpitiya wind corridor as the Envision Final EIA (Sep-2025) reference project, which is used here only as an **ex-post corroboration**, never as a driver.

2. Wind resource & siting assessment

This section is built from a **fresh ERA5 retrieval run this session** (single-point ARCO timeseries, reproducible fixed 2005–2024 window) plus the committed bankable-AEP loss chain that feeds the finance model.

2.1 Fresh long-term resource (ERA5 ARCO, 2005–2024, 175,320 h, 100 % coverage)

Reference period	Mean hub WS (150 m)	Net AEP P50	Capacity factor
2020–2024 (recent 5-yr)	7.294 m/s	523.0 GWh	0.3742
2015–2024 (current decade)	7.317 m/s	527.5 GWh	0.3773
2005–2024 (long-term 20-yr)	7.460 m/s	551.2 GWh	0.3943
P90 (20-yr basis)	—	480.0 GWh	—

Interannual coefficient of variation: **2.69 %** (low — a stable resource). Coverage complete (175,320 / 175,320 hours). Single-cell retrieval → spatial representativeness recorded as **not assessed** (honest caveat; a neighbourhood GIS sample is the next step for micro-siting representativeness).

2.2 ★ Material finding — a statistically significant secular-stilling trend

The 20-year series carries a **downward wind-speed trend that is significant at the 5 % level**:

- **Mann-Kendall** $\tau = -0.453$, **p = 0.0047** (significant)
- **Sen's slope** = -0.219 m/s per decade (95 % CI $-0.335 \dots -0.087$)
- **OLS** slope -0.210 m/s/decade, $R^2 = 0.365$
- Decade means: 2000s **7.575** → 2010s **7.485** → 2020s **7.294** m/s
- Classification: **secular trend / stilling** — a sustained decline with material explained variance.

Bankability consequence: a forward P50 should be weighted toward the **current-climate period**, not the full 20-yr mean. The assessment therefore recommends the **recent-5-yr basis: P50 523.0 GWh (CF 37.4 %)** as the central bankable figure, with the 20-yr 551.2 GWh as the upside. This is the honest, IEC-61400-15-1 / MEASNET-consistent treatment (long-term reference, regime-shift-adjusted).

2.3 Committed bankable AEP used by the finance model

The finance run uses a **more conservative, fully-loss-adjusted** bankable AEP (from the committed bankable_aep chain), not the raw ARCO figure:

- ERA5-fitted Weibull **A = 8.199, k = 2.665**, mean **7.29 m/s** (this equals the fresh recent-5-yr mean of 7.294 — direct corroboration), air density 1.15 kg/m³.
- Loss stack **14.48 %**: PyWake Bastankhah-Porté-Agel wake **7.28 %** (15-turbine layout; TurbOPark bound 8.93 %), availability 97 %, electrical 2 %, curtailment 2 %, other/environmental 1 %.
- IEC 61400-15-2 pre-construction **over-prediction haircut 2.0 %** (a conservative policy choice, not derived from this project's own EYA).
- → **Net P50 464.3 GWh, CF 0.332; P90 404.4 GWh**.
- Ex-post cross-check: the Envision EN220/10.0 Final EIA reports **P50 464.5 GWh** — within **~0.04 %** of the model output (corroboration only).

Reconciliation (fresh vs committed): the fresh ARCO recent-5-yr figure (523.0 GWh, CF 0.374) is a resource-level net; the committed 464.3 GWh (CF 0.332) additionally carries the full IEC loss stack + the 2 % over-prediction haircut — i.e. the finance model runs **~11 % below** the raw resource, deliberately. The two are consistent (same 7.29 m/s wind climate); the model uses the conservative number.

2.4 Turbine & layout

IEA Wind Task 37 **10 MW reference** (198 m rotor, 345 W/m², low-specific-power for the 7.3 m/s site), hub 150 m, cut-in 3 / rated 11 / cut-out 25 m/s. Open reference curve (not OEM-certified) — a stand-in for a future low-wind 10 MW OEM machine. Layout: 15 turbines, linear north-south coastal, avg spacing 650 m (3.8 D); IEC spacing compliance is marginal (6 compliant / 3 marginal / 6 violating pairs) — a micro-siting optimisation item, and the reason wake is modelled granularly rather than assumed flat.

2.5 Wind rose (fresh ARCO directional distribution, 12 × 30° sectors)

Sector	0° N	30°	60°	90° E	120°	150°	180° S	210°	240°	270° W	300°	330°
Frequency	4.3 %	13.1 %	13.2 %	3.8 %	1.1 %	1.4 %	4.4 %	31.8 %	22.7 %	1.5 %	1.0 %	1.6 %

Prevailing sector 210° (SW). The climate is **bimodal monsoon**: a dominant **SW band (210–240°)** at **~54.5 %** (the SW monsoon) plus a secondary **NE band (30–60°)** at **~26.3 %** (the NE monsoon return). Provenance: single-cell ERA5 (~0.25°), indicative of the prevailing sector, not mast-validated on-site — a met-mast directional campaign is a bankability item. This directional structure is what makes the committed **linear north-south layout efficient** — the turbine string sits broadside to the dominant SW flow, minimising in-line wake.

2.6 Micro-siting (TopFarm on PyWake — fresh candidate this run)

A boundary-, spacing- and exclusion-constrained AEP maximisation (DTU TopFarm on the PyWake Bastankhah-Porté-Agel model), seeded with the committed 15-turbine layout and driven by the fresh ARCO wind rose:

	AEP (wake-included)	Note
Committed baseline (linear N-S, 650 m)	551.0 GWh	scored on the PyWake model at the ARCO Weibull
Optimised candidate	558.5 GWh	3.0-D min spacing respected (594 m)
Uplift	+7.47 GWh (+1.36 %)	KPI-neutral candidate — informs layout, not the finance case

Read: the committed layout is already well-matched to the SW-dominant rose; a re-sited candidate captures a **modest ~1.4 %** additional energy by further trimming wake. This is a candidate for the micro-siting workstream, not a change to the headline economics (which use the committed 464.3 GWh bankable AEP).

2.7 GeoGIS spatial screen (fresh GeoTIFF export)

A native-resolution ERA5 GIS export (3×3 cells at 0.25° , bilinear-downscaled to 0.05°) rendered WS150 / capacity-factor / AEP-per-turbine GeoTIFFs and a spatial-representativeness verdict over the ~80 km neighbourhood:

Metric	Value	Tolerance
Site-cell mean WS150	7.294 m/s	—
Neighbourhood (9-cell) mean WS150	7.344 m/s	—
Site-cell deviation from neighbourhood	−0.69 %	± 8 % ✓
Neighbourhood spatial spread	28.3 %	15 % ✗
Representative verdict	False	—

Read: the site cell sits essentially *at* the local mean (−0.7 %, comfortably within tolerance), so the reanalysis is a sound central estimate — but the wider ~80 km neighbourhood is spatially **heterogeneous** (28 % spread, a coastal/inland resource gradient), which trips the representativeness flag. The honest conclusion: the P50 level is defensible, but the regional variability argues for an **on-site met-mast campaign** before financial close, not reliance on single-cell reanalysis. Artifacts: `artifacts/gis/` (ws150/CF/AEP GeoTIFFs, coarse + fine, each with a provenance sidecar) + DataLake manifest.

3. Technical basis

Item	Value
Rated capacity	159.6 MW (15×10.638 MW)
Rotor / hub	198 m / 150 m
Power curve	IEA Reference 10 MW (IEC 61400-12-1:2022, open reference)
Air density correction	site 1.15 vs ref 1.225 kg/m ³
Degradation	0.5 %/yr (industry-standard onshore aging)
Wake model	PyWake Bastankhah-Porté-Agel (granular, 7.28 %)

4. Capex & opex

4.1 Capex — bottom-up WBS (\$159.6 M base = \$1,000/kW, AACE Class 3)

WBS line	USD	Share
Turbines (WTG supply, CIF + inland)	110,124,000	69.0 %
Foundations	7,660,800	4.8 %
Grid connection (220 kV line)	7,660,800	4.8 %
Turbine erection	5,107,200	3.2 %
Array collection (33 kV)	5,107,200	3.2 %
Substation bay	5,107,200	3.2 %
Development / owner's / land	5,586,000	3.5 %
Site preparation	3,064,320	1.9 %
Financing fees	3,192,000	2.0 %
Contingency (~1.5 %)	2,394,000	1.5 %
Project management	2,553,600	1.6 %
Site development	2,042,880	1.3 %

WBS line	USD	Share
Base capex total	159,600,000	100 %

\$/kW benchmark: \$1,000/kW sits between the SINOHYDRO Mannar EPC quote (~\$889/kW) and the Envision Dutch Bay EIA (~\$1,420/kW), and near the IRENA-2024 global onshore benchmark (\$1,041/kW).

Import-levy uplift (prudent posture, #738): PAL 5 % + import-SSCL 2.5 % **paid** on the 0.69 imported share = + **\$8.2593 M** duties (customs import duty on wind gensets is FREE, HS 8502.31). Capex VAT is relieved via the BOI §17 / bonded route. → **Gross financed capex \$167.86 M (\$1,051.75/kW)**. Duties capitalise into the depreciable base and flow to IDC, gearing and NPV.

4.2 Opex (\$3.0 M/yr = \$20/kW-yr)

O&M contract \$2.0 M · insurance \$0.5 M · land lease \$0.3 M · admin/SPV \$0.2 M. Unrecoverable 18 % input VAT on O&M is paid (IPP supply is VAT-exempt → no input credit). Opex held flat-nominal (no USD escalation applied; LKR cost erosion enters via the FX curve).

5. Financing structure

Parameter	Value
Debt sizing	dual-DSCR auto-sculpt to a 1.30 target, capped at 70 % gearing
Realised gearing	0.355 (auto-solved down to hold the covenant)
Max senior debt	\$59.59 M
Tenor / grace	15 yr / 2-yr interest-only
Tranche mix	LKR-local 45 % · USD-DFI 10 % · USD-commercial 45 %
Tranche rates	LKR 13.39 % (UIP-implied: USD 7.5 % + 5.89 % drift) · DFI 6.5 % · USD-comm 7.5 % (blended kd ~7.63 %)
Credit-support fees	DFI guarantee 75 bps + PRI 100 bps per yr on outstanding senior debt (sized inside the sculpt)
DSRA	6 months
Balloon	~48.9 % at maturity, cash-sweep resolution (breaches the modelled 10 % refinance-risk covenant — a structuring flag)
IDC	\$10.75 M

The LKR tranche at the **UIP-implied 13.39 %** (not a concessional 8 %) is the honest market-rate assumption pending a real term sheet; it is the single biggest lever pushing the equity return negative. The ~49 % balloon under a cash-sweep is a genuine structuring weakness to resolve before close.

5.1 Capital-structure optimisation — financing cannot rescue the tariff

A capital-structure optimiser swept **36 debt-mix / gearing candidates** to maximise equity IRR. **All 36 are negative.** The best (DFI 40 % / USD-commercial 60 % / no LKR) lifts equity IRR from the committed **−7.85 % to −6.21 %** (min-DSCR 1.30, LLCR 1.17); the worst (LKR-heavy) is **−8.18 %**. So the *optimal* financing structure gains only ~1.6 pp and **still cannot make equity positive** — a quantitative proof, from the finance side, that the binding constraint is the flat-LKR tariff, not the capital structure. It corroborates the Sobol result (§7.5): the tariff, not the debt mix, is what moves the return.

6. Returns

6.1 Headline KPIs (live canonical run — reproduces the 5th-gen canon byte-identically)

KPI	Value
Project IRR (unlevered, real)	-0.001166233356501311 (-0.12 %)
Equity IRR (levered)	-0.07853839579881527 (-7.85 %)
Project NPV @ WACC	-\$79,273,039
Equity NPV @ ke	-\$81,592,659
Min DSCR (covenant fold / per-period)	1.3 / 1.30
Avg / max DSCR	1.39 / 2.57
LLCR / PLCR	1.268 / 1.307
Equity MOIC	0.411 (< 1 — sponsors do not recover their outlay)
WACC (build-up) / project discount	~9.94 % / 10.02 %
Total CFADS (20-yr)	\$166.08 M
Total equity distributed	\$51.78 M

6.2 The equity bridge (why leverage makes it worse, not better)

Unlevered project IRR -0.12 % → levered equity IRR -7.85 %. Because the **asset return (-0.12 %)** is far below the **blended cost of debt (~7.6 %)**, leverage is **dilutive**, not accretive: every borrowed dollar earns less than it costs, and the FX-inflating USD/blended debt service on flat-LKR revenue amplifies the loss. Add the SL 15 % dividend WHT, the UIP LKR rate, credit-support fees, import levies and the COD-aligned operating FX, and the equity return sinks below -7 %. The re-baseline chain (documented in Appendix A) shows each step.

7. Sensitivity & risk

7.1 Scenario suite (8 live runs) — no scenario clears a positive NPV

Scenario	Project IRR	Equity IRR	Min DSCR	Project NPV	LLCR
Optimistic	7.40 %	+4.90 %	1.06	-\$22.71 M	1.20
Base case	4.20 %	-2.04 %	0.61	-\$51.36 M	0.94
Equity case	4.20 %	+0.25 %	1.40	-\$51.36 M	1.64
Pessimistic	2.23 %	-5.67 %	0.41	-\$70.04 M	0.81
Capex lean (SINOHYDRO)	1.32 %	-6.52 %	1.30	-\$74.96 M	1.29
Hybrid wind+solar	0.31 %	-4.39 %	1.30	-\$107.33 M	1.49
Lender (canonical)	-0.12 %	-7.85 %	1.30	-\$91.81 M	1.28
Capex prudent (EIA)	-1.99 %	-9.44 %	1.30	-\$144.75 M	1.37
Solar-only	-4.73 %	-8.80 %	1.30	-\$111.53 M	1.56

Every scenario has a negative project NPV. The best case (optimistic) reaches +4.90 % equity IRR but still a -\$22.71 M NPV — i.e. even the upside does not clear the 12 % equity hurdle. Only two of the nine frames (optimistic, equity case) clear a positive equity IRR at all. The result is robust to the technology and cost frame; the binding constraint is the tariff, not the capex.

7.2 Monte-Carlo distribution (live run)

An LHS Monte Carlo over six correlated drivers (capacity factor, tariff, opex, capex, FX, incremental curtailment; **Iman-Conover** rank correlation — capex↔opex +0.35, CF↔curtailment +0.20, FX left uncorrelated because the flat-LKR PPA breaks the revenue-FX link). Run at **2,500 trials** (100 % success, 0 failed iterations, **0 toy-fallback substitutions**). The run's own convergence trace confirms the distribution is settled: the 95 % CI half-width on mean equity IRR narrows monotonically to **±0.12 pp** (±0.0007 on mean min-DSCR) by trial 2,500:

Metric	P10	P50	P90	Mean
Equity IRR	-13.0 %	-9.1 %	-5.0 %	-9.1 %
Project IRR	-3.6 %	-1.0 %	+1.8 %	-1.0 %
Project NPV	-\$124.7 M	-\$97.5 M	-\$69.4 M	-\$97.4 M
Min DSCR	1.257	1.291	1.300	1.285
LLCR	1.264	1.278	1.296	1.279

The negative-equity verdict is robust, not a point estimate: under the COD-aligned FX path equity IRR is **negative across effectively the entire distribution** — the whole-sample maximum across 2,500 trials is just **+0.15 %**, and **project NPV is negative in 100 % of trials** (best trial -\$38.3 M). Meanwhile **min-DSCR stays ≥ 1.217 throughout** — the *debt* is safe across the distribution; it is the *equity* that is underwater. (VaR/CVaR + covenant-breach detail in [artifacts/capital_risk_report.html](#); NPV-distribution chart below.)

7.3 Downside production

P90 (1-yr) net AEP **404.4 GWh** vs P50 464.3 GWh (P90/P50 ≈ 0.871). The dual-DSCR sizer binds on the P50; the P99 downside factor (0.80 of P50 ≈ P90 territory) is what constrains debt capacity.

7.4 AEP sensitivity (fresh tornado)

A fresh AEP tornado on the committed resource basis (net-modelled base **473.84 GWh**, before the 2 % haircut → 464.3 bankable):

Driver	Swing	AEP range (GWh)
Wind-speed bias ±5 %	±20.5 %	424.6 – 521.5
Power curve (IEA vs DTU 10 MW)	-16.0 %	398.3
Shear exponent ±0.04	±6.6 %	458.1 – 489.6
Losses ±20 %	±6.4 %	458.8 – 489.2

Wind-speed uncertainty dominates — roughly 3× the next driver. Combined with the significant secular-stilling trend (§2.2), this is exactly why the pre-construction P50 haircut and the P90 debt-sizing are the load-bearing conservatism in the model, and why an on-site measurement campaign (§2.7) is the highest-value pre-close derisking step.

7.5 Global sensitivity (Sobol / PAWN / Morris)

A variance-based global sensitivity analysis (SALib), over the same six drivers, for **equity IRR**:

Driver	Sobol total-order (ST)
Tariff (LKR/kWh)	0.47
Capex	0.19
FX (LKR/USD)	0.12
Capacity factor	0.12
Curtailment	0.06
Opex	0.04

The flat-LKR tariff is quantitatively the #1 equity-IRR driver (ST 0.47 — larger than the next two combined). This is the finance-side counterpart to the AEP tornado (§7.4, where wind-speed dominates *energy*): once the energy is financed, the *administered tariff* dominates the *return*. It is the analytical confirmation of the whole thesis — the binding lever is the revenue line, and it is fixed. (Morris μ^* ranks capex first on a one-at-a-time basis; Sobol total-order, which captures interactions, puts tariff first.)

8. Grid & interconnection (advisory screen — fired end-to-end)

The in-house grid-strength screen (analytics.grid #870 stack; **pandapower 3.3.0 / andes 2.0.0 / opendssdirect 0.9.4** all resolved) was **run this pass** with `grid.study_enabled=true`, and is **KPI-neutral** — the finance KPIs are byte-identical to canon with it on. Results:

- **SCR@POC $\approx$ 0.94 ("weak")** by pandapower IEC-60909 (fault level 150.1 MVA at the 33 kV POC vs 159.6 MVA plant) → **GFL at risk; grid-forming (GFM) likely required** (consistent with the NSO round mandating GFM). (!) **Screening estimate**, `bankable:false` — the 150 MVA is driven by a *placeholder* 6 Ω connection reactance, not a CEB/NSCC-issued fault level. A bankable connection study (PSS®E / PowerFactory vs the utility base case) is a condition precedent.
 - **RMS ride-through screen** ran all three cases (LVRT / HVRT / frequency); verdicts are the honest None because the ANDES dynamic solve is opt-in (the default is RMS-envelope only). The IEEE-519 harmonic screen ran. (The `poc_envelope / freq_response` sub-screens report not-applicable for a single-tech wind scenario — expected, not a failure.)
 - **Provenance / verification discipline:** running the full stack surfaced — and this session fixed — **two real grid/GIS bugs**: a rasterio `boundary_clip` segfault under GDAL 3.10.3 (fixed, merged #929) and a `ride_through` kwarg `TypeError` that had silently degraded the screen on every run (fixed, PR #930). Both are exactly the defects a "run every module" pass is meant to catch.
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9. The honest economics

- **Why equity IRR is negative:** a flat-nominal-LKR tariff with no escalation and no FX index, against a cost base that is ~69 % USD-denominated (turbines) and debt that is ~55 % USD/blended, financed while the LKR depreciates ~5.9 %/yr. Revenue is fixed in a currency that loses ~6 %/yr of USD value; costs and debt service are not. The scissors close on the equity.
 - **The "cheap USD debt" illusion:** the USD-DFI 6.5 % / USD-commercial 7.5 % tranches *look* cheaper than the 13.39 % LKR tranche, but on flat-LKR revenue their effective LKR cost rises with depreciation — which is exactly why the UIP-implied LKR rate (13.39 %) is the correct comparator and why leverage is dilutive here.
 - **The counter-case:** the project's Final EIA reports an **18.07 % economic IRR (EIRR)**. That is not a contradiction — it is a *different frame*: EIRR counts economy-wide benefits (avoided fossil generation, capacity value) under a concessional/blended capital stack, whereas this study's financial eqIRR is the sponsor's after-tax cash return at the administered flat-LKR tariff. **The project is economically attractive to Sri Lanka and financially value-destructive to a merchant sponsor** — the classic case for blended finance / concessional capital or a restructured tariff.
 - **Merchant-tail caveat:** there is no merchant upside to rescue the tail — the PPA is flat for 20 years, so the downside (FX, curtailment, resource stilling) is one-sided.
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10. ESG / EIA context & development readiness

Environmental & social: Final EIA approved (Envision, Sep-2025); E&S management plan in place → **GREEN**. Marine/coastal setting (Dutch Bay); IFC-PS / EP4-aligned. No standalone EP4 Climate Change Risk Assessment yet commissioned (honest gap; the resource-stilling finding in §2.2 is a physical-climate risk that a CCRA should formalise).

Development-readiness register (roll-up: RED):

Workstream	Status	Note
Environmental & social	[fired] Green	Final EIA approved
Land	[not fired] Amber	Site identified; lease in progress
Permits	[not fired] Amber	Generation licence + approvals under application
Grid connection	[not fired] Amber	220 kV line separate CEB-funded project
PPA	[not fired] Amber	CEB standardized template; tariff not executed
EPC	[not fired] Amber	SINOHYDRO bottom-up quote; contract not awarded
Financing	Red	equity IRR negative at flat-LKR tariff; debt+equity uncommitted

11. Recommendation with conditions precedent

Conditional NO-GO at the current flat-LKR 20.3 LKR/kWh tariff. The project is value-destructive on a merchant/standalone basis in every scenario run. Do not reach financial close as structured.

Path to a conditional GO — pursue at least one, ideally a combination, of:

1. **Revenue restructuring (highest impact):** tariff FX-indexation or CPI escalation, or a USD-denominated / USD-linked PPA. This attacks the root cause (flat-LKR vs USD costs).
2. **Concessional / blended capital:** a materially concessional senior or first-loss/grant tranche that lowers the blended cost of debt below the asset return — the frame under which the EIA's 18.07 % EIRR is bankable.
3. **Debt-structure fixes:** resolve the ~49 % balloon (longer tenor, amortising structure, or committed refinance) and secure a real LKR term sheet (the 13.39 % UIP rate is an assumption, not a quote).
4. **Grid & resource DD:** a bankable PSS®E/PowerFactory grid-connection study (replacing the screening estimate) and a met-mast-validated resource campaign that formalises the secular-stilling haircut.
5. **Commercial DD:** execute the PPA (tariff currently assumption-tier), award the EPC, and close the land/permit items.

Until at least the revenue or the capital stack is restructured, the honest position is that DutchBay is an **economically valuable, financially unbankable** project — a candidate for concessional/blended finance, not commercial project finance.

Appendix A — Assumptions & evidence register (provenance tiers)

Assumption	Value	Source	Tier
Tariff	20.3 LKR/kWh flat	CEB standardized PPA (not executed)	assumption
Debt terms	15-yr, blended kd ~7.63 %, DSCR-sculpted	indicative (no term sheet)	assumption
Capex	\$1,000/kW bottom-up WBS	NREL SAM/LandBOSSE; IRENA-2024 anchor; AACE Class 3	benchmark
Opex	\$20/kW-yr	IRENA RPGC 2024 / NREL ATB 2024-25	benchmark
Capacity factor	0.332 net P50	ERA5→PyWake→power-curve→AEP, -2 % haircut	derived
FX	333.79 LKR/USD, 5.89 % depr	CBSL anchor / FRED / BIS; 2026-Q2 pinned	measured
Tax	CIT 30 %, div WHT 15 %, no holiday	SL IRA, IMF-reformed 2024-26	benchmark
Degradation	0.5 %/yr	industry-standard onshore	benchmark
Discount rate	~9.9 % build-up WACC	modelled capital structure	derived
Climate risk	analyst screen	no formal EP4 CCRA yet	assumption

Re-baseline chain to the current canon (honest audit trail): FX 300→333.79 · ERA5-fitted Weibull (AEP 483.6→473.8) · project-discount construction-lag fix · degradation 0→0.5 %/yr · FX-drift 3 %→5.89 % · 2 % AEP over-prediction haircut (473.8→464.3) · non-statutory levy removal · SL dividend WHT + IDC capitalisation · UIP LKR debt rate 8 %→13.39 % · credit-support fees (guarantee 75 bps + PRI 100 bps) · import levies + indirect tax (SSCL-on-revenue reversed; PAL+import-SSCL paid) · **F5-01 COD-aligned operating FX** (#1034/#1038: parametric operating and hedge curves advance by the two-period construction window; capex, levy and IDC tax basis stay at the 333.79 financial-close spot). Net: projIRR → -0.12 %, eqIRR → -7.85 %, NPV → -\$91.81 M, gearing → 0.355.

Appendix B — Full scenario KPI table

See §7.1 (8 live scenarios). Source: `_out/feas924/suite/<scenario>/.../kpis.json`, all engine v15.3.1 / SHA 7e64d33 (regenerated 2026-08-17).

Appendix C — Monte-Carlo method

100k-trial LHS (seed 42); drivers {capacity_factor, tariff, opex, capex, fx, curtailment} with uniform supports base-(1 ± 2·rel_std); Iman-Conover correlation (capex↔opex +0.35, CF↔curtailment +0.20). VaR/CVaR at 95 %; covenant floors DSCR 1.20 / LLCR 1.25. Lender-grade capital-risk report rendered at 2000 bounded trials (≥ the 1000-trial floor). Artifacts: `capital_risk_report.html`, `npv_distribution_equity_npv.png`.

Appendix D — Reproduce recipe & provenance

```
Repo: ~/Downloads/dutchbay-epc-model @ origin/main 7e64d336759292d1fa1c3f1533f6ec20ea6c0250
Engine: v15.3.1 Python 3.11 (.venv) [finance re-baselined 2026-08-17, F5-01 #1034/#1038]
Canon: .venv/bin/python run_full_pipeline_v14.py config=scenarios/dutchbay_lendercase_2025Q4.yaml
       → project_irr -0.001166233356501311 · equity_irr -0.07853839579881527 · min_dscr 1.3 (byte-
       identical to the F5-01 canon)
Wind: ERA5_REQUEST_CONFIG=artifacts/era5_request_dutchbay.yaml .venv/bin/python -m
       wind_resource.era5_retrieval
       → ARCO single-point 2005-2024, P50 523.0 (recent-5yr) / 551.2 (20-yr) GWh; wind_rose (prevailing
       210° SW); secular-stilling MK p=0.0047
Tornado: analytics.wind.aep_tornado.tornado_from_config(lender) → wind-speed ±20.5% dominant
       driver
Micrsit: wind_resource.layout_optimizer.optimize_layout (TopFarm/PyWake) → +1.36% AEP uplift candidate
       (baseline 551.0 → 558.5 GWh)
GIS: GIS_EXPORT_CONFIG=wind_resource/config/gis_export_dutchbay.yaml -m analytics.gis.gis_export
       → WS150/CF/AEP GeoTIFFs (coarse 0.25° + fine 0.05°) + spatial-representativeness (False; 28.3%
       neighbourhood spread)
Suite: run_full_pipeline_v14.py config=scenarios/
       dutchbay_{basecase,equitycase,optimistic,pessimistic,capex_*,hybrid_windsolar,solar_only}_2025Q4.yaml
Emitters: +emit_capital_risk_report / +emit_tech_comparison / +emit_interaction_grid /
       emit_executive_workbook (all four fired)
FreshAEP: analytics.wind.aep_summary_builder.build_aep_summary_from_config(lender) → net P50 464.36 GWh /
       CF 0.3322 (= committed 464.3 → BASELINE reproduced the long way)
MC: analytics.mc.engine.run_monte_carlo_analysis(lender, n_trials=2500, seed=42) [+ a full 100k
       cross-check]
       → equity_irr P10/P50/P90 -13.0/-9.1/-5.0%; project_npv negative in 100% of trials; min_dscr
       ≥1.217 (2500 ≈ 100k, converged)
Sobol: analytics/cli/cli_sensitivity_hydra.py (SALib Sobol+PAWN+Morris) → equity-IRR ST: tariff 0.47
       (dominant) > capex 0.19 > fx 0.12
Optimiz: analytics/cli/cli_capital_structure_optimize_hydra.py → 36 debt-mix candidates, ALL negative;
       best -6.21% (DFI40/USD60)
Grid: run_full_pipeline_v14.py config=<lender with grid.study_enabled:true> +emit_grid_screen=true
       → SCR@POC 0.94 (weak, pandapower IEC-60909); ride-through 3 cases (advisory); KPI byte-identical
       (neutral)
```

Full-stack note. The finance engine drives production off `project.capacity_factor` and reads a *committed* `resource.aep_summary_path`; it does NOT re-run the wind stack. This study fired the wind/GIS/micro-siting stack **separately** and confirmed it reproduces the committed AEP (464.36 vs 464.3 GWh) — so the baseline holds

computed the long way. Two data/terrain GIS layers remain **honestly blocked** (need external Global-Wind-Atlas / Copernicus-DEM / ESA-WorldCover rasters, absent locally); they are flagged, not fabricated.

Verification discipline: the canonical lender run reproduced the 5th-generation canon **byte-identically** (project_irr / equity_irr / min_dscr to 1e-9); the grid screen is advisory and KPI-neutral (byte-identical whether on or off); every figure in this study traces to a live artifact in `artifacts/`. Where an input is `assumption-tier` (tariff, debt terms), it is labelled as such — this is a development-stage feasibility, not a financed model.

Prepared on the live DutchBay EPC model, 2026-07-07. Honest-economics basis: the project is technically viable and economically valuable to Sri Lanka, but financially value-destructive to a merchant sponsor at the committed flat-LKR tariff. Bankability requires a restructured revenue or concessional capital stack.